

Supplementary Material

Task-Oriented Semantic Granularity Selection for Bandwidth-Constrained V2V Cooperative Perception

Products. Every number in this document is read from the same committed products as the main paper: results/main/replay_summary.csv, results/main/fixed_references.csv, results/main/tau_feasible.csv, results/main/true_e2e_ap.csv, results/main/action_distribution.csv, results/main/feature_importance_frozen.csv, results/main/step4_collaboration_harm.csv, results/sensitivity/feature_ablation.csv, results/sensitivity/difficulty_frozen.csv, results/sensitivity/collaborator_scale.csv, results/sensitivity/delivery_semantics_bracket.csv, results/sensitivity/robustness_frozen.csv, results/sensitivity/r23_scene_bootstrap.csv, results/sensitivity/r23_object_message_bler.csv, results/sensitivity/r23_fragmentation_harq.csv, results/baselines/two_gate.csv, results/baselines/two_gate_dgate.csv, results/baselines/importance_map_jscf/two_regime_edge_clean.csv, results/p4b/replay_summary.csv, results/p4b/perclass_ELF.csv, results/manifests/FROZEN_MANIFEST.json and results/channel_bler_sionna.csv.

This document carries material moved out of the main paper for length. It is not a second set of results: the two appendices below are the prior-protocol arms labelled as such in the main text, and the cue table is the definition list of the 21 committed ego-side cues. Section and label names are preserved so that the main text’s pointers resolve against this document.

I. Ego-Side Online Task Cues: full definition list

The selector’s task-cue vector $z_t \in \mathbb{R}^{21}$ is defined in full below; the main paper lists five representative cues and points here.

II. When Are Perception Cues Necessary? LDPC Cliff versus JSCC Graceful Degradation

Protocol note. A prior-protocol arm, not re-evaluated under the frozen P2 protocol: its selector is trained and evaluated inside the codec-comparison pipeline, sharing the retired engine’s BLER lookup and 200-realisation CSI convention. Its numbers may not appear in the same sentence as a frozen-replay number, and no main-text claim rests on them.

Under LDPC + QAM a one-line threshold tracks the learned selector (the main paper); does that equivalence

TABLE I
Ego-side online task cues used by the CA-TOSG selector ($d = 21$). All cues are computed locally on the ego vehicle before message selection, removing the causality issue that would arise from depending on the collaborator’s state.

Category	Cue	Definition
Agent context	num_cavs	number of CAVs in scene
	ego_num_objects	ego-detected objects
Frame shape	ego_origin_lidar_shape_0/1	raw LiDAR tensor shape
Range stats	pcd_num_points	total point count
	pcd_mean_range	mean LiDAR range
	pcd_max_range	max LiDAR range
	pcd_std_range	range std. dev.
Distance bins	pcd_near_20m	points within 20 m
	pcd_mid_20_50m	points 20–50 m
	pcd_far_50_80m	points 50–80 m
	pcd_very_far_80m	points beyond 80 m
Quadrant cnt.	pcd_front_points	front-half points
	pcd_back_points	rear-half points
	pcd_left_points	left-half points
	pcd_right_points	right-half points
Long-range	pcd_front_far_30m	front points > 30 m
	pcd_front_far_50m	front points > 50 m
Density bins	pcd_density_0_20	density 0–20 m
	pcd_density_20_50	density 20–50 m
	pcd_density_50_80	density 50–80 m

survive a feature codec without a sharp SNR cliff? We replace the feature branch with importance-map JSCC [1] and re-run the per-frame pipeline: per-frame JSCC F1 is re-cached at each SNR (1,000 frames per SNR, AWGN and Rayleigh), the $\{L, \text{Feature-JSCC}\}$ oracle relabelled and the selector retrained—no perception-backbone change.

The mechanism is a codec property. The LDPC + QAM feature F1 cliffs over the 8–12 dB decoding transition (onset 8.0 dB), so SNR is highly informative; the learned JSCC feature F1 is essentially flat at ≈ 0.89 across 0–20 dB, so the estimated SNR says almost nothing about whether a feature message will help on a given frame.

This flips the comparison (Table II). Both codecs are evaluated on the same 2,170 test frames under the identical 200-seed protocol, separating (a) whether the graceful-channel gain exists and resides in the cues—estimated leakage-free in-distribution by 5-fold cross-validation, every frame scored by a selector that never saw it and τ tuned on the training folds only—from (b) whether a deployed, validate-frozen selector realises it, under the headline cross-split protocol.

(a) In-distribution. Under LDPC + QAM the AWGN

edge over the best SNR threshold is only +0.002 F1 (95% CI [+0.001, +0.003]): small, and—as the feature ablation shows—not attributable to the perception cues alone (the main paper).¹ Under JSCC the best SNR threshold collapses to the Fixed- L operating point—it cannot gate on an uninformative SNR—and the cue-based edge, measured by the in-distribution k -fold estimator, widens to +0.022 F1 ([+0.020, +0.025]) under AWGN, +0.018 ([+0.016, +0.020]) under Rayleigh, and +0.020 ([+0.018, +0.022]) under OFDM on the test split, and to +0.012–+0.015 on the Culver-City domain shift (all CIs exclude zero). In the separate 200-realisation held-out comparison on validate frames, the cue-based selector recovers 56–62% of the clairvoyant oracle’s headroom over Fixed L across the three channels. The two quantities are distinct and are reported separately: under AWGN the oracle headroom is 0.0291 F1 and the selector recovers +0.0181 of it, with 0.0275 and +0.0158 under Rayleigh and 0.0281 and +0.0158 under OFDM. In-distribution, then, the graceful-channel decision is content-bound and carried by the ego-side cues—a gain no SNR threshold reaches on these frames, and one that (b) below shows does not survive the cross-split move.

(b) Cross-split deployment. The signal does not transfer: deployed on test the frozen selector over-selects the feature action (JSCC C-request rate 0.14 $\rightarrow$ 0.42) and falls just below Fixed- L —edge -0.004 (AWGN), -0.008 (Rayleigh), -0.005 (OFDM) on test, ≈ 0 on Culver-City. The gain is present but not portable; capturing it under cross-domain deployment, e.g. by online or self-supervised adaptation [2], is future work.

Whether task-oriented cues are needed is determined by the feature codec’s channel response. Under a cliff codec the decision is channel-bound and SNR suffices; under a graceful codec it becomes content-bound and the ego-side cues—which predict whether object-level detection is already adequate—become the operative signal, in-distribution. Closing the cross-split transfer gap (b) is the remaining challenge.

III. Boundary of the Method: a Second Detection Backbone

Convention. This arm was trained and evaluated on the full-collaborator caches, i.e. before the corrigendum of Change-log P0: it is internally consistent (its own budget, its own N_{cw} , its own references) and its relative conclusion stands, but its absolute F1 and payload figures are on

¹This in-distribution per-frame edge (+0.002) is distinct from two other LDPC-regime quantities: the feature-ablation gain of the cues (Full vs. channel-state-only, +0.0090, 95% CI [+0.0089, +0.0091], the main paper) and the matched-payload margin over the retuned threshold (+0.0032, the main paper). The three measure the selector along the cue, the equal-bandwidth, and the per-frame axis respectively. Under the corrected single-collaborator convention all three are positive, with the ordering $+0.002 < +0.0032 < +0.0090$: the cue axis, smallest and indistinguishable from zero under the retired full-collaborator accounting (-0.0002), is now the largest of the three — weakening the single message widened the gap the cues have to close, and they close part of it.

TABLE II

Learned-selector edge over the best SNR-threshold rule on the same 2,170 test frames (200-seed protocol), reported two ways. In-dist. edge: leakage-free 5-fold cross-validation on the evaluation split (each frame scored by a selector that never saw it, τ tuned on the training folds only). Deployed edge: selector trained on validate and frozen, applied cross-split (the headline protocol). In-distribution the JSCC edge is 9–11 $\times$ the LDPC-cliff edge—an SNR threshold cannot reach it—while under LDPC the margin is the selector’s function form, not a cue gain (cues add no significant F1, the main paper). The deployed JSCC edge is negative: the validate-trained selector does not transfer the graceful-channel gain across the domain shift (Appendix II(b)).

Feature codec	Channel	In-dist. edge	Deployed edge
LDPC + QAM (cliff)	AWGN	+0.002 [+0.001, +0.003]	-0.002 [-0.003 , -0.001]
Importance-map JSCC	AWGN	+0.022 [+0.020, +0.025]	-0.004 [-0.007 , -0.001]
Importance-map JSCC	Rayleigh	+0.018 [+0.016, +0.020]	-0.008 [-0.011 , -0.006]
Importance-map JSCC	OFDM	+0.020 [+0.018, +0.022]	-0.005 [-0.008 , -0.003]

a different accounting basis from the single-collaborator main experiment and are not tabulated beside it.

Positioning. This appendix reports a limitation, not a generality result. It is placed here because its conclusion is negative: the selection procedure developed in the main text is in-sample effective on a second backbone but does not transfer to held-out splits under the equal-budget protocol. No claim in the main text depends on it.

Every result above uses one perception backbone. To test whether the selection procedure—not the particular features—is what carries the behaviour, we repeat the whole pipeline on a second detector, SECOND [3], under an equal-budget controlled comparison: the feature message is charged the same per-frame channel budget as the mainline ($B_F \equiv 0.99$ Msym, the same N_{cw} and the same BLER tables), so the two backbones are compared at equal channel cost. We do not claim that SECOND’s feature tensor compresses to that size, and we declare no codec for it.²

Both SECOND checkpoints are the official OpenCOOD ones, and we verify them before use rather than trusting the file: after a lossless axis-order conversion required by the sponcv version in our environment, official inference reproduces the published AP@0.7 to +0.0002 on both splits for the late-fusion model (0.7752 vs. 0.775; 0.6822 vs. 0.682) and to within +0.0019 for the intermediate model. The three per-frame branches (E , L , F) are then re-derived with the same scorer, the same canonical union ground truth and the same IoU-0.5 convention as the mainline, and the entire downstream pipeline—grid expansion, scene-level 9-fold LOSO, the budget walk, and the 200-realisation paired replay—is re-run with the mainline code, each stage first verified to reproduce its own committed mainline product bit-for-bit.

The result is that CA-TOSG is in-sample effective but does not transfer under the equal-budget protocol. Table III gives the paired comparison against an SNR-

²The transmitted tensor was measured, not assumed: SECOND’s cross-link BEV tensor is 6,758,400 elements per collaborator before compression and 352,000 at the auto-encoder bottleneck. Both are recorded as measurements; neither is used as the operative payload under the equal-budget convention.

TABLE III

Second backbone (SECOND), equal-budget controlled. Paired $\Delta F1$ of the frozen selector against the SNR-threshold rule tuned to the same budget, over the same 200 CSI samplings; 95% frame-level paired bootstrap intervals. Positive favours CA-TOSG. Second-backbone arm, not deployed; descriptive, no decision is taken from this table.

Split	$B_{\max} = 0.10$	$B_{\max} = 0.20$	$B_{\max} = 0.30$
Validate (in-sample)	+0.0058 [+0.0057, +0.0059]	+0.0095 [+0.0094, +0.0096]	+0.0081 [+0.0081, +0.0082]
OPV2V test	-0.0014 [-0.0015, -0.0013]	-0.0047 [-0.0048, -0.0046]	-0.0066 [-0.0067, -0.0065]
Culver-City	-0.0029 [-0.0030, -0.0028]	-0.0117 [-0.0119, -0.0115]	-0.0167 [-0.0169, -0.0165]

threshold rule tuned on validate for the same target budget. On validate—the split the selector is trained on—it is ahead at every budget (+0.0058 to +0.0095 F1). On both held-out splits it is behind at every budget, with all confidence intervals excluding zero.

Two mechanisms account for the gap, and both are visible in the arm’s own numbers. First, the L - F margin is narrower on this backbone: SECOND’s object-level branch scores 0.869 / 0.879 / 0.879 mean per-frame F1 on validate / test / Culver-City against a compressed-feature branch at 0.897 / 0.904 / 0.939, so there is simply less to gain from choosing correctly, and the same selection error costs proportionally more. Second, the budget walk selects a different candidate family: all three frozen SECOND selectors carry `class_weight=balanced`, whereas all three mainline selectors carry `class_weight=None`. The reweighted family fits the in-sample class mix well—agreement with the oracle is 0.833 on validate—but transfers poorly, falling to 0.578 on test and 0.534 on Culver-City, where the selector degenerates to near-always- L (E recall 0.027 on test and 0.000 on Culver-City). The E -collapse of Section IV-G is therefore worse on this backbone, not milder: under Rayleigh the selector’s ρ_E is 0.002 (test) and 0.000 (Culver-City) while the oracle spends 0.157 and 0.133. The large payload reductions this arm shows (67–90%) follow from that collapse to L and are not evidence of good selection.

We report this as a boundary of the present method rather than a tuning target: nothing was retrained and no threshold was adjusted after seeing these numbers.

IV. Material moved from the main paper

The four subsections below are reproduced verbatim from the main paper’s results section; the main text keeps a short summary of each and points here.

A. Collaboration Is Not Always Beneficial

Collaboration is not unconditionally beneficial, and that is what makes an explicit per-frame selector necessary rather than a convenience. Two failure modes call for two responses. When the channel cannot carry a feature message, requesting one spends the collaborator’s transmission budget for nothing and collapses the output

to the ego-only floor (the ego vehicle’s own pre-fusion detection); the design answer is already in place – the oracle removes any action whose frame-level failure probability is ≥ 0.999 from its feasible set (the same 0.999 constant as the feasibility mask of the main paper), and on failure the pipeline reverts to the ego-only output rather than a phantom feature. When the channel can carry the message, requesting it can still cost accuracy: on the easy stratum the selector’s realised output falls below even the always-object-level (Fixed- L) baseline.³ This mode has no masking answer; its remedy is left to future work. Two GSV-verified quantifiers bound where the ego-side harm sits: the ego-only output strictly exceeds the object-level fused output on 1.5 / 5.8 / 0.2% of validation / test / Culver-City frames, and – from the same per-frame (comp – ego) identity, re-derived on the corrected single-collaborator caches – the compressed-feature message, when delivered, yields lower frame F1 than the ego-only fallback on 1.3 / 6.5 / 0.9% of frames. Test carries the harm most, consistent with fusion having the least to add in thin scenes (mean 15.2 ground-truth objects on test vs 27.8 on validate and 41.0 on Culver-City). No new signalling is needed to act on this: E (ego-only, no cooperative perception payload) is already a deployed action of $\mathcal{S}$ with its own codepoint, so the remedy is a selection-policy question—the frozen selectors almost never choose it (Section IV-G)—not a message-format one.

B. Where2comm as an Adjacent-Technology Reference

Where2comm is an adjacent-technology reference, not a baseline: it is not ranked against any arm of this paper, and no claim here rests on it. To place CA-TOSG beside the most prominent spatial-selection cooperative perception method, we report reference-only measurements of Where2comm [4] on the same OPV2V validate split (1,980 frames). Three caveats bound what it can support, and they are stated rather than buried: the numbers come from our own 50-epoch reproduction, not the authors’ released model; they are measured under a perfect channel, so no part of this paper’s transport model applies to them; and they were scored by the retired global-sort scorer, which is a different track from the frozen AP chain that produces every headline number here. Our reproduction reaches $AP@0.5 = 0.871$ under a perfect channel, and we deliberately draw no ranking from it. It is not comparable to any number in this paper, on three independent axes at once: a different scorer (the retired global-sort track, not the frozen AP chain), a

³Test Easy stratum (top tercile of the ego’s own object-level F1, the §IV-H stratification), evaluated under a deterministic reliable-channel condition (AWGN 16 dB; frame-level BLER ≈ 0 , well above the 8.0 dB onset), isolating the difficulty axis from channel variability: the frozen selector’s realised F1 at $B_{\max} = 0.20$ is 0.9749 vs the Fixed- L baseline 0.9796 – a gain of -0.0047 (frame-level paired 95% CI [-0.0074, -0.0024]; $n = 713$ frames). The selector requests F on 241 of these 713 frames; on the remaining 472, where it requests L , its output is frame-identical to Fixed- L , so the paired difference arises entirely on the F -request frames – the loss is a structural consequence of requesting features on already-easy frames.

perfect channel rather than the modelled transport, and—decisively—a different collaborator convention, since the reproduction fuses every collaborator while the corrected main experiment pays for one. The direction of such a comparison is not even stable: under the corrected accounting the validate Fixed- L reference is 0.7819, below 0.871, which reverses the ordering an earlier version of this section reported. Numbers measured under three different conventions cannot be ordered, and we do not order them.

The comparison is also orthogonal to CA-TOSG: Where2comm is a spatial-mask method that decides which feature regions to send within a fixed feature-level granularity, whereas CA-TOSG decides whether to send feature-level information at all and at what granularity, given the channel. The two compose: Where2comm can be deployed inside the C_q branches of CA-TOSG as an alternative feature codec, while the granularity selector decides when feature-level communication is worth its payload and channel risk. Under a bandwidth-constrained, time-varying link, even a feature-level message is realised only when the channel can deliver it—precisely the regime CA-TOSG arbitrates.

C. Collaborator Scale

The evaluation so far treats the collaborator set as given. To ask how the policy behaves as more vehicles are reachable, we sweep the number of requested collaborators $N \in \{1, 2, 3\}$, keeping the frozen selector, λ^* and τ^* untouched.

N is not the operative quantity: what matters is the effective count k_{eff} , since requesting three cannot conjure a third. Of the 1,980 validate frames, 733 are saturated at $N = 1$, 1,081 at $N = 2$ and 1,234 at $N = 3$; on Culver-City the supply runs out, so $N = 2$ and $N = 3$ are the same arm. Comparisons are conditioned on the same frames.

F1 then rises with k_{eff} with sharply diminishing returns. At $B_{\text{max}} = 0.20$ on validate, $N = 1 \rightarrow 2$ buys +0.0300 F1 for +0.1183 Msym and the third adds +0.0061 for +0.0991 Msym—a fifth of the gain for three-quarters of the channel; on test the pattern is flatter (+0.0096 then +0.0008). The payload sub-linearity is a supply effect: the action distribution is unchanged across N , so the extra channel use comes only from frames that gained a collaborator.

Because “request N ” is ambiguous when fewer than N collaborators exist, we bracket the two delivery semantics—charging only for what is delivered, versus charging for what was requested—on validate at $N = 2$, on the 964 frames where the two differ. The bracket is tight: charging for what is delivered rather than for what was requested raises realised F1 by +0.00016, +0.00036 and +0.00099 across the three budgets, at identical payload—the request is what is charged under both—so the conclusions here do not depend on which convention is adopted. Finally, the budget does not transfer automatically. The selector is frozen against an average per-frame budget measured with a single collaborator; total communication

grows with the number of collaborators actually served, so the original budget guarantee does not carry over unchanged. This is measurable rather than hypothetical: on validate at $B_{\text{max}} = 0.20$, $N = 3$ spends 0.36842 Msym per frame, well above the 0.20 it was frozen to satisfy — and under the corrected single-collaborator accounting even $N = 2$ (at 0.26937) already breaches it. A deployment that serves more collaborators must therefore re-freeze, or budget on k_{eff} rather than per collaborator.

D. Deployment Robustness and Cost

Four practical imperfections—a noisy SNR estimate, channel-state aging, a stale request, and a misread channel type—degrade the selector only gracefully, because an uncertain channel state falls back to L (Table IV): SNR-estimation noise up to $\sigma \approx 1$ dB costs ≤ 0.0002 F1; a Jakes model at 60 km/h decorrelates the estimate within ~ 10 ms, costing -0.0025 before plateauing; a one-frame-stale decision costs at most -0.0109 under our i.i.d.-per-frame assumption (changing 18.2% of decisions). The near-immunity to estimation noise follows from the channel-type-anchored policy of Section IV-F: perturbing continuous $\hat{\gamma}_t$ rarely flips a decision anchored on the discrete channel-type flag, and under Rayleigh the action remains L regardless. Finally, the deployed Random Forest runs in 52.1 ± 5.6 ms per frame (P95 = 58.3 ms) on a single CPU core. Selector-only latency is 52.1 ms; end-to-end system latency is not measured here. The figure is the slowest of the three frozen selectors over 1,000 batch-1 trials each; the deployment claim must hold for every budget, so the worst case is the one quoted.

Misreading the channel type. The selector takes the channel-type indicator c_t as an input, and c_t carries the single largest Gini share of any feature, so its reliability is a deployment question rather than a detail. We flip c_t on a random fraction of frames and replay the frozen policy (P3 sensitivity arm, channel_misclassification.csv). The degradation is graceful and roughly linear in the flip rate: on test at $B_{\text{max}} = 0.20$ the arm’s own no-flip baseline is 0.89701 realised F1, falling to 0.89620 at a 5% flip rate, 0.89542 at 10% and 0.89384 at 20%, while the payload remains nearly unchanged (0.14036, 0.14033, 0.14032 and 0.14051 Msym). The asymmetry of the two error directions is what keeps the cost small. A frame misread as Rayleigh biases the selector towards the conservative L , which is safe and merely forgoes a gain. A frame misread as AWGN may lead the selector to request F ; the block error probability is then evaluated on the true channel, so the message is likely to be lost and the frame falls back to the ego-only output rather than to L . Only the second direction can lose accuracy, and it loses it to the ego-only floor, not to a corrupted feature map. The arm is a sensitivity replay, not a deployed configuration: its no-flip cell is not the mainline replay’s.

A fourth imperfection concerns the cheap message rather than the expensive one. The utilities above model the object-level link as reliable ($\text{BLER}_L = 0$),

TABLE IV

Deployment robustness (OPV2V test; realised frame F1). Each row perturbs one imperfection while holding the others ideal; losses are bounded because an uncertain channel state defaults to the robust L message. Re-derived under the single-collaborator convention with the frozen selectors at $B_{\max} = 0.20$ (results/sensitivity/robustness_frozen.csv); the CSI-noise and Jakes models are the ones used previously, so only the selector, grid and utilities changed.

Imperfection	Operating point	$\Delta F1$
SNR-estimation noise σ	≤ 1 dB	-0.0002
	2 dB	-0.0006
	5 dB	-0.0021
CSI aging (Jakes, 60 km/h)	stale by ~ 10 ms	-0.0025 (then plateaus)
Request delay	1-frame-stale decision	-0.0109 (pessimistic)

which is an assumption about the low-rate path, not a measurement. Re-weighting the object-level utility analytically as $\text{eff}'_L = \text{late}(1 - \text{BLER}_L) + \text{ego BLER}_L$ over $\text{BLER}_L \in \{0.01, 0.05, 0.10\}$, with every policy frozen and blind to it, leaves the payload — and therefore the payload advantage — untouched by construction, and costs F1 roughly linearly: on test at $B_{\max} = 0.20$, $\text{BLER}_L = 0.10$ costs the selector 0.00634 F1, the threshold rule 0.00588 and Fixed L 0.00734 (results/sensitivity/r23_object_message_bler.csv). The selector’s margin over Fixed L therefore widens as the cheap link degrades, while its gap to the threshold rule widens against it, from -0.00010 to -0.00056 — the threshold rule leans harder on the feature action ($\rho_F = 0.20$ against 0.12) and is correspondingly less exposed to an unreliable L . We report this rather than the favourable half alone; it remains inside the pre-registered margin of 0.005 at every BLER_L evaluated.

E. Selector Training

The selector is a Random Forest classifier [5] over the 23-dimensional input $x_t = [z_t, \hat{\gamma}_t, c_t]$, with labels the channel-aware oracle s_t^* of the oracle definition of the main paper. Selection is budget-indexed and pre-registered rather than hand-tuned: a fixed block of 112 candidates (hyperparameter settings $\times$ a payload-penalty grid λ) is evaluated once under scene-level 9-fold leave-one-scene-out cross-validation on validate—each of the nine scenes held out exactly once, 1,008 fold rows—and for each budget $B_{\max} \in \{0.10, 0.20, 0.30\}$ the candidate list is walked in descending out-of-fold F1 until one meets the budget, which freezes that budget’s (λ^*, τ^*) and hyperparameters. The three frozen selectors are $\lambda^* = 0.05/0.02/0.00$ with $\tau^* = 18/12/8$ dB at $B_{\max} = 0.10/0.20/0.30$; all three carry class_weight=None—the balanced candidates were walked past for exceeding their budget—and FROZEN_MANIFEST.json is the authority for every one of these values. Each frozen configuration is then refitted once on the full 1,980-frame validate split and frozen—never retrained—for every reported test and Culver-City result. Run-to-run variance is exposed not by a within-validate resplit but by averaging all realised-F1 and

payload metrics over 200 Monte-Carlo channel realisations per operating point (per-frame SNR $\sim \mathcal{U}[0, 20]$ dB and channel type $\sim \text{Bernoulli}(0.5)$ AWGN/Rayleigh).

F. Selector Decisions and Feature Importance

Fig. 1 shows the per-mode selection ratios ρ_s against SNR, with the frozen selector overlaid on that budget’s payload-penalised oracle and the stacked-area panel showing the share of each action of $\mathcal{S} = \{E, L, F\}$ at every SNR. Under AWGN the selector’s adaptation knee is at 10 dB, where the frame-error cliff clears: ρ_F steps from 0 to 0.472 on validate (0.433 on test, 0.160 on Culver-City) and is flat above it, with ρ_L falling correspondingly. Under Rayleigh ρ_L stays at essentially 1 across the whole range. The panel also makes the E -collapse of Section IV-G visible: under Rayleigh the oracle spends a substantial share on the ego-only action— $\rho_E = 0.157$ on test and 0.133 on Culver-City—while the frozen selector’s ρ_E is 0.002 and 0.000. E is a deployed action that the imitation objective almost never selects, which is a limitation of the trained policy rather than of the action set. It is, however, not a limitation a simpler policy repairs: a hand rule with an explicit E gate, fitted over three difficulty proxies in both orientations, selects E in no split $\times$ budget cell ($\rho_E = 0.00000$ throughout, the main paper) — on these cues E is structurally out of reach of a monotone threshold, and the frozen selector’s small but non-zero ρ_E is more than any of them attains. Under the corrected single-collaborator convention this limitation became cheaper without becoming milder. The behaviour is unchanged — the selector still almost never chooses E — but the price of those missed calls fell from $0.58\text{--}0.61\delta$ to $0.44\text{--}0.47\delta$ on test, where $\delta = 0.005$ is the pre-registered non-inferiority margin. The oracle’s own appetite for E moved in opposite directions on the two splits: the number of grid cells it labels E rose on validate ($335 \rightarrow 529$) and fell on test ($6,303 \rightarrow 5,696$), so the correction does not simply make ego-only more attractive everywhere. Feature importance.

the feature-importance table of the main paper report the top Gini feature importances of the deployed selector. The two channel-side features dominate: the channel-type indicator c_t contributes 34.2% of importance and the estimated SNR $\hat{\gamma}_t$ a further 27.5%, so two of the 23 inputs carry 61.7% between them against 38.3% spread over the 21 perception-side cues. The strongest of those, pcd_mean_range, reaches only 3.0%. Two cautions belong with that number. Importance is not sufficiency: a Gini share says how often a feature is split on, not that the remaining 38.3% is redundant, and the feature ablation shows the opposite — with the channel features alone the selector stops requesting features at the tighter budgets and never recovers the full selector’s operating point (the main paper). And the split is transport-specific: it is measured under an LDPC + QAM cliff where the channel state is nearly decisive by construction; under graceful degradation the same cues carry the decision instead (the

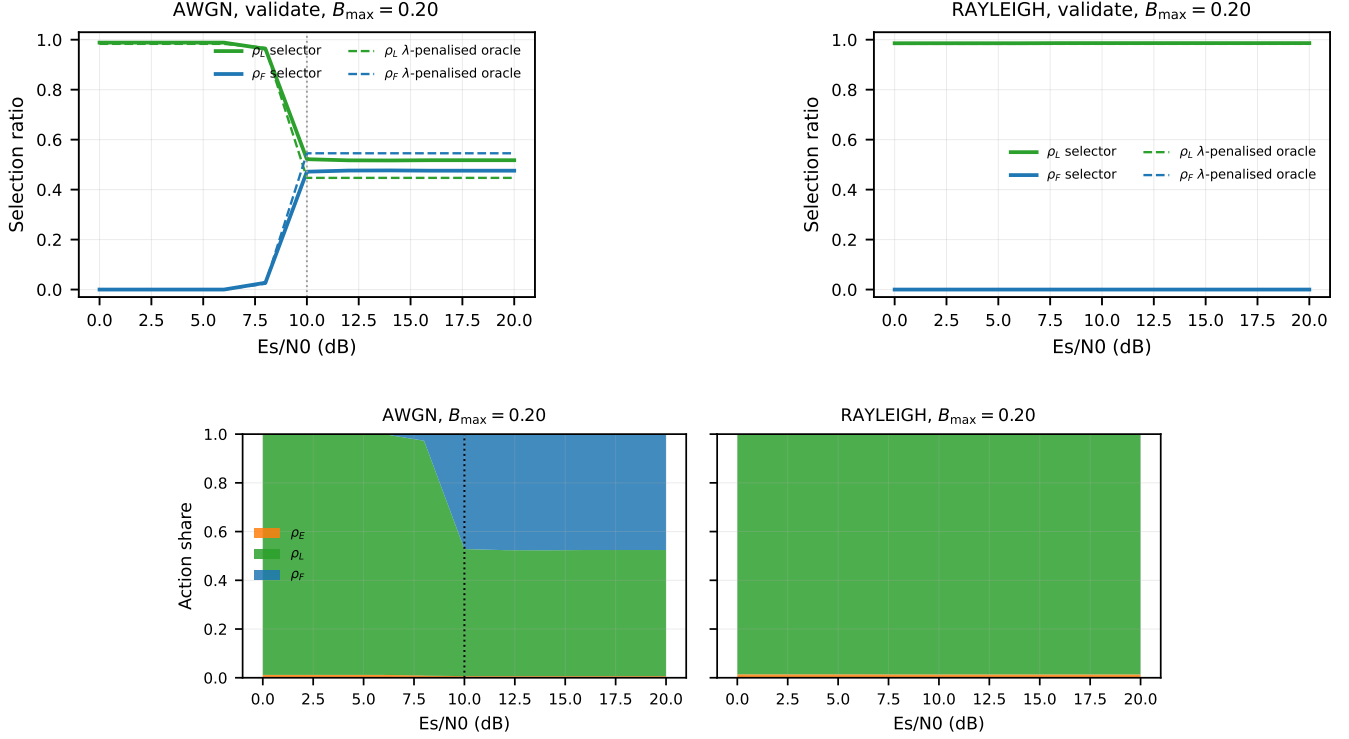


Fig. 1. Per-mode selection ratios versus SNR on OPV2V validate at $B_{\max} = 0.20$, from the frozen grid. Top: the frozen selector (solid) against that budget's λ -penalised oracle (dashed) — the oracle sees the frame BLER, not the per-frame block outcome, so it is not clairvoyant. Bottom: stacked-area view of the selector's action distribution over the deployed action set $\mathcal{S} = \{E, L, F\}$ under AWGN and Rayleigh (the non-deployed C_{256} physical-layer comparator); the dotted line marks the measured 10 dB policy knee. Two readings: under AWGN ρ_F steps from 0 to 0.472 at 10 dB and is flat above it, and under Rayleigh the selector's ρ_E is 0.002 (test) / 0.000 (Culver-City) while the oracle spends $\rho_E = 0.157$ / 0.133 there — the E -collapse limitation of Section IV-G. The three-budget version is the per-budget action distributions (repository figure product).

supplementary material). That the channel-type indicator c_t outranks the estimated SNR $\hat{\gamma}_t$ is consistent with the frame-level channel physics: under Rayleigh fading the feasible set collapses to L regardless of SNR, so the channel-type flag is the single highest-information split for the decision.

G. Communication–Perception Pareto Frontier

The headline table compares policies at a single operating point. To show that CA-TOSG sits on the attainable payload–F1 frontier rather than merely better on average, we sweep the Lagrangian selector $s_t^* = \arg \max_s (F_t^s - \lambda B_s)$ over λ and plot every policy on the payload–F1 plane (Fig. 2). As λ grows from 0 (max F1) to large values (min payload), the oracle traces the achievable upper frontier; the fixed feature-level policies C_{16} and C_{256} sit far below it (high payload, low realised F1 due to the LDPC cliff), i.e. they are strictly dominated. CA-TOSG and a λ -cost RF retrained on $\arg \max_s (F_t^s - \lambda B_s)$ labels lie on the frontier between Fixed L and the oracle. The P2 product is three frozen selectors, one per budget, so the operating point is budget-indexed: on test they realise F1 0.89148, 0.89691 and 0.89783 at 0.037, 0.141 and 0.21196 Msym/frame for $B_{\max} = 0.10, 0.20$ and 0.30 . At $B_{\max} = 0.20$ that payload is 14.3% of the feature-level payload B_F .

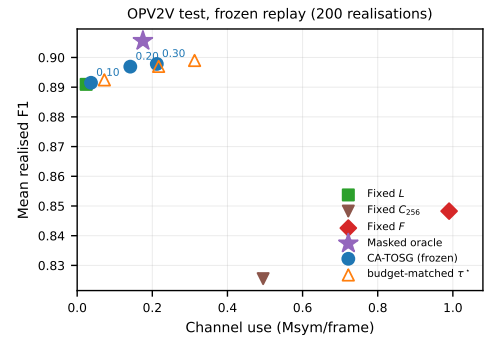


Fig. 2. Communication–perception plane on OPV2V test, from the frozen 200-realisation replay. The axis spans the full range so Fixed F (0.99 Msym) is visible rather than clipped. The three frozen selectors are shown budget-indexed (circles, labelled by $B_{\max}$) beside the SNR-threshold rule tuned to the same budget (open triangles); the feasibility-masked oracle is a separate marker (star). Fixed F and Fixed C_{256} are dominated by Fixed L at far higher channel use.

H. Where the Gain Concentrates: Difficulty Stratification

A channel-averaged mean masks where CA-TOSG matters. We stratify frames into easy/medium/hard terciles by the ego's own object-level detection F1 (low = hard) and evaluate the frozen selectors at a single reliable-

TABLE V

Selector feature ablation and the SNR-threshold challenger (train: validate, eval: frozen OPV2V test at $B_{\max} = 0.20$, $T = 2,170$ frames, 200 SNR/channel realisations). Each variant is retrained through the identical selection pipeline—same candidate set, same scene-level 9-fold LOSO, same frozen walk under the budget—with only the feature columns changed. Neither half of the input alone produces a graded policy: both the channel-only and the cues-only variants collapse to always transmitting L at this budget (feature-request rate $\rho_F = 0$), so only the full 23-feature selector reaches an intermediate operating point. The cues-plus- $\hat{\gamma}$ cut of the earlier ablation was not re-run under the corrected protocol and is therefore not listed.

Selector	#feat	Realised F1	Deployed payload (Msym)
Channel only ($\hat{\gamma}, c$)	2	0.89095	0.02400
Perception cues only	21	0.89092	0.02392
Full (all features)	23	0.89691	0.14141
SNR-threshold (τ^*)	—	0.89701	0.21679
Fixed L	—	0.89095	0.02400

channel operating point—deterministic AWGN at 16 dB, one point of the pre-registered SNR grid rather than a channel draw (the difficulty stratification (repository figure product)). At $B_{\max} = 0.20$ the CA-TOSG-over-Fixed- L gain rises monotonically with difficulty: on test it is -0.0047 (95% CI $[-0.0074, -0.0024]$) on easy frames, $+0.0056$ ($[+0.0027, +0.0085]$) on medium frames and $+0.0660$ ($[+0.0591, +0.0730]$) on hard frames. Validate has the same shape ($+0.0058/+0.0441/+0.0470$); Culver-City is weaker throughout ($+0.0011/+0.0021/+0.0310$). The value of CA-TOSG is therefore two-dimensional: it materialises where a hard frame meets a reliable channel, and a channel-averaged headline necessarily dilutes it. On easy frames the deployed selector over-requests F on test, and the frozen product puts the cost at -0.00471 F1 (95% CI $[-0.00742, -0.00236]$) at $B_{\max} = 0.20$, growing monotonically with the budget $-0.00020, -0.00471, -0.01129$ across $0.10/0.20/0.30$ —as the looser cap lets more easy frames be sent as F . The effect is split-dependent rather than universal: on validate and Culver-City the same stratum gains at every budget ($+0.00035$ to $+0.01016$ and 0.00000 to $+0.00558$), so it is a property of the test split’s easy tercile, where fixed L already reaches 0.97962 and there is almost nothing left to buy. A payload-penalised ($\lambda > 0$) operating point on the Lagrangian frontier of Section IV-G would mitigate it.⁴

V. Feature-ablation comparison table

The table the ablation section of the main paper points to.

VI. Tables moved from the main paper

The two tables the main paper points to; its prose carries their conclusions and numbers.

⁴The channel-averaged companion view reported in an earlier version of this subsection is withdrawn rather than recomputed: it was a product of the retired 200-realisation policy engine and its selector, not of the frozen selectors evaluated here, and the two are different quantities that may not be placed side by side.

TABLE VI

Notation for the semantic granularity levels. F and C_{16} denote the same message; the C_q form is used only where the modulation order is the subject.

Symbol	Meaning	Channel use	Role
E	ego-only, no transmission	$B_E = 0$	deployed action; also the deli
L	object-level message	$B_L = 0.024$	deployed action
$F \equiv C_{16}$	feature-level, 16-QAM	$B_F = 0.99$	deployed action
C_{256}	feature-level, 256-QAM	0.495	physical-layer comparator, no

TABLE VII

Pre-registered sensitivity analyses (test at $B_{\max} = 0.20$ unless stated). Products: r23_scene_bootstrap.csv, r23_object_message_bler.csv, r23_fragmentation_harq.csv, r25_fragmentation_replay.csv.

Check	Quantity	Result
Scene-level bootstrap	$\Delta F1$ 95% CI	$[-0.001508, +0.001522]$ (18.938×)
	ΔB 95% CI	$[-0.119307, -0.029856]$ (45.177×)
BLER _L = 0.10	F1 cost, selector	-0.00634 (threshold -0.00588 , F
	payload	unchanged by construction
Fragmentation + HARQ	AWGN 8 dB BLER	$0.402400 \rightarrow 0.100363$ ($k=2$), 0.05
	payload factor	$1.226954\times, 1.120770\times$
	Rayleigh 0–20 dB	frame BLER 1.0 for every k

VII. True end-to-end AP verification: full protocol and readings

The verification prose the main paper compresses to two sentences.

A. True End-to-end AP Verification

The realised frame F1 reported so far is computed under the analytical block-loss model of Eq. (4) of the main paper, where a lost feature block falls back to the ego vehicle’s own pre-fusion detection. To confirm that this analytical F1 proxy translates to actual end-to-end detection AP under a deployed pipeline, we run a true end-to-end inference in which: (i) the deployed selector picks s_t per frame from $(\hat{\gamma}_t, c_t)$; (ii) cached per-frame predictions from the OPV2V late-fusion checkpoint (for L) and the attentive-compression checkpoint (for C_q) are concatenated according to the selector’s choice; (iii) each C_q frame is dropped with probability $\text{BLER}_q(\gamma_t, c_t)$ and falls back to the ego-only (pre-fusion) prediction; (iv) the resulting box set is scored against ground-truth boxes with the OpenCOOD VOC-style AP computation at IoU thresholds $\{0.3, 0.5, 0.7\}$, averaged over the 200 paired stochastic channel realisations of the frozen replay. The per-SNR breakdown is the committed product true_e2e_ap_by_snr.csv.

Two observations support the central claim. First, the boundary of the feature-active regime is measured, not stipulated: reading the frozen selector’s feature-request rate off the pre-registered SNR grid, ρ_F is zero at every AWGN point up to 8 dB and steps to its plateau at 10 dB on all three splits—0.472 on validate, 0.433 on test and 0.160 on Culver-City—which is exactly where the 16-QAM frame-error cliff clears (Section III-C of the

TABLE VIII

Generalisation of the validate-trained frozen selectors (no retraining) to the OPV2V test ($T = 2,170$ frames, scene-disjoint) and Culver-City ($T = 550$ frames, real-road-layout domain shift) splits. Every row is the mean over the same 200 paired CSI samplings of the frozen replay (SNR $\sim \mathcal{U}[0, 20]$ dB, 50/50 AWGN/Rayleigh); each frozen selector is paired with the SNR-threshold rule tuned to the same budget. Share is channel use as a fraction of Fixed F (0.990 Msym).

Policy	Mean F1	Channel use (Msym)	Share of Fixed F
OPV2V test ($T = 2,170$)			
Fixed L	0.8910	0.024	2.4%
Fixed F (LDPC + 16-QAM)	0.8483	0.990	100.0%
Fixed C_{256} (LDPC + 256-QAM)	0.8255	0.495	50.0%
CA-TOSG $B_{\max}=0.10 / \tau^*$	0.89148 / 0.89247	0.0368 / 0.0724	3.7%/7.3%
CA-TOSG $B_{\max}=0.20 / \tau^*$	0.89691 / 0.89701	0.1414 / 0.2168	14.3%/21.9%
CA-TOSG $B_{\max}=0.30 / \tau^*$	0.89783 / 0.89900	0.2120 / 0.3125	21.4%/31.6%
Channel-aware oracle (masked)	0.9056	0.175	17.7%
OPV2V Culver-City ($T = 550$)			
Fixed L	0.8466	0.024	2.4%
Fixed F (LDPC + 16-QAM)	0.8099	0.990	100.0%
Fixed C_{256} (LDPC + 256-QAM)	0.7793	0.495	50.0%
CA-TOSG $B_{\max}=0.10 / \tau^*$	0.84667 / 0.84956	0.0244 / 0.0718	2.5%/7.2%
CA-TOSG $B_{\max}=0.20 / \tau^*$	0.84975 / 0.85858	0.0675 / 0.2174	6.8%/22.0%
CA-TOSG $B_{\max}=0.30 / \tau^*$	0.85932 / 0.86316	0.1823 / 0.3146	18.4%/31.8%
Channel-aware oracle (masked)	0.8643	0.266	26.8%

main paper).⁵ At that boundary the true end-to-end AP on validate is 0.8287 at IoU 0.5 and 0.7292 at IoU 0.7, and it is flat above it. The selector reaches this while paying 0.300 Msym/frame averaged over the feature-active regime instead of the 0.990 of always-on Fixed F —a 70% saving—though on validate it recovers only part of the perfect-channel reference of the underlying attentive-compression model (0.8369 on validate, Section VI-A of the main paper) rather than essentially all of it, as an earlier version of this section claimed. Second, the Rayleigh AP curve is flat across the full SNR range and matches Fixed L , confirming that the selector’s conservative fading-regime behaviour, identified in Section VI-A of the main paper, is not a frame-F1 artefact but a genuine end-to-end property.

VIII. Figure and table moved from the main paper (R41)

The difficulty/importance figure and the feature-importance table; the main paper carries their conclusions and numbers.

IX. Common-Volume Diagnostic: How Much of the Headroom Is Field of View?

Motivation. The branches of this paper are separate trained pipelines, and one of the four ways they differ is the configured perception range: the late-fusion checkpoint behind L covers $x \in [-70.4, 70.4]$ m, the compression checkpoint behind F covers $x \in [-140.8, 140.8]$ m. Both are scored against one canonical ground-truth set, which extends to $|x| \approx 119$ m. Ground truth beyond the L

⁵Validate shows a 0.4% toe at 8 dB, the single grid point at which the frame BLER is neither ≈ 0 nor ≈ 1 (0.402); test and Culver-City are exactly zero there. We therefore take 10 dB as the regime boundary on all three splits and report the toe rather than smoothing it away. Earlier versions of this section stipulated ≥ 14 dB; that threshold was not derived from any measurement and is withdrawn.

TABLE IX

Top-7 features by Gini importance for the deployed CA-TOSG selector (trained on OPV2V validate, $T = 1,980$ frames). The two channel-side features account for 61.7% of total importance; the 21 perception-side cues carry the remaining 38.3%. Read directly from the frozen model, results/main/feature_importance_frozen.csv.

Feature	Gini importance
channel_is_rayleigh	0.342
est_snr_db	0.275
pcd_mean_range	0.030
pcd_left_points	0.028
ego_num_objects	0.025
pcd_std_range	0.023
pcd_front_far_30m	0.021
$\hat{\gamma}_t + c_t$ subtotal	0.617

branch’s range is therefore a structural miss for L and a detectable object for F , so part of the measured L – F gap is field of view rather than semantic granularity. The size of that part was unknown, so we measured it.

Method. Every prediction and every ground-truth box, from the same committed caches the headline table uses, is cropped to the intersection volume $|x| \leq 70.4$, $|y| \leq 40$ m by box centre, and the unchanged scorer is re-run. Two caveats belong with the numbers: the crop is centre-based, so a box straddling the boundary is kept or dropped whole; and the CA-TOSG rows use 20 of the 200 paired realisations, the three fixed references being deterministic. The percentage of canonical ground truth removed is 18.89% on validate, 9.94% on test and 20.48% on Culver-City.

Reading. Every arm’s absolute AP rises inside the common volume, which is expected: the removed ground truth is ground truth no branch could reach. What matters is the gap. On validate the headroom falls from 0.0550 to 0.0117, on Culver-City from 0.0970 to 0.0252, and on test it changes sign, from 0.0240 to -0.0061 —inside a shared

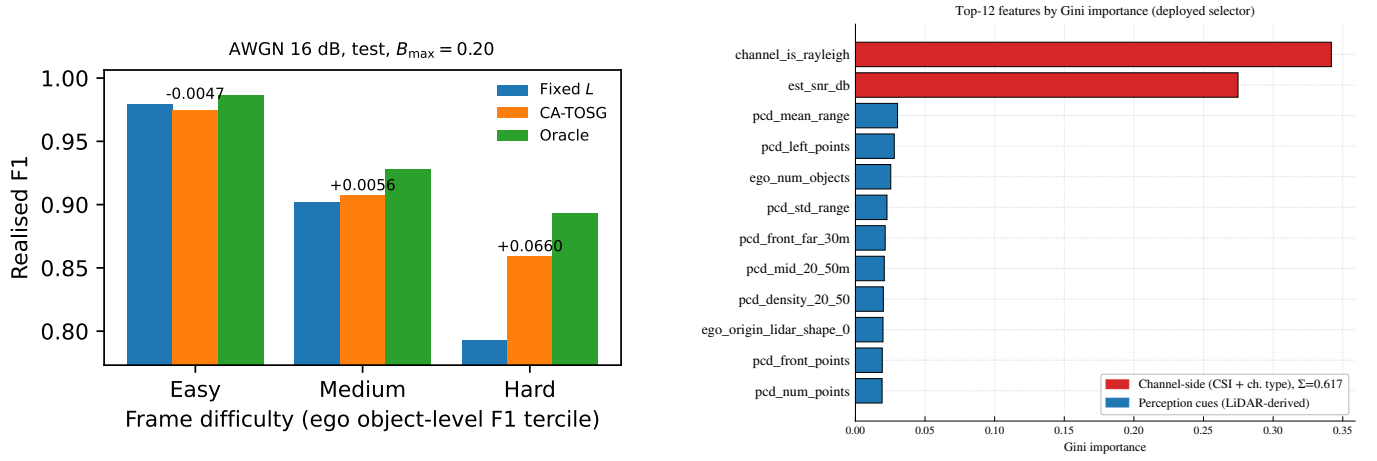


Fig. 3. Left: realised F1 by frame-difficulty tercile (ego object-level F1) on OPV2V test at $B_{\max} = 0.20$ at a single reliable-channel grid point (AWGN 16 dB) with the frozen selector; numbers above bars are CA-TOSG–Fixed- L , the hard-frame gain being $+0.0660$ (95% CI $[+0.0591, +0.0730]$). Right: top-12 features by Gini importance of the frozen $B_{\max} = 0.20$ selector—the channel-side features jointly account for 61.7%, the 21 perception-side cues for the remaining 38.3%, none above 3.0% alone. Importance is not sufficiency (the ablation section of the main paper).

TABLE X

Common-volume diagnostic, AP@0.5. Deployed is the frozen table (each branch as configured); common volume re-scores the same cached outputs inside $|x| \leq 70.4$, $|y| \leq 40$ m. The headroom row is the feature ceiling minus fixed L . Source: results/diagnostics/common_volume_ap.csv.

Split	Policy	Deployed	Common volume	Δ
validate	ego-only	0.6116	0.7539	+0.1423
	Fixed L	0.7819	0.9064	+0.1245
	Feature ceiling	0.8369	0.9181	+0.0812
	headroom	0.0550	0.0117	−0.0433
test	ego-only	0.7350	0.8162	+0.0812
	Fixed L	0.8691	0.9429	+0.0738
	Feature ceiling	0.8931	0.9368	+0.0437
	headroom	0.0240	−0.0061	−0.0301
Culver-City	ego-only	0.6185	0.7775	+0.1590
	Fixed L	0.7299	0.8825	+0.1526
	Feature ceiling	0.8269	0.9077	+0.0808
	headroom	0.0970	0.0252	−0.0718

X. Channel-averaged comparison table moved from the main paper (R42)

The main paper reads this table in two sentences; the full row set is here.

References

- [1] Y. Sheng, H. Ye, L. Liang, S. Jin, and G. Y. Li, “Semantic communication for cooperative perception based on importance map,” *Journal of the Franklin Institute*, vol. 361, no. 6, p. 106739, 2024.
- [2] C. Liu, J. Chen, Y. Chen, R. Payton, M. Riley, and S.-H. Yang, “Self-supervised adaptive weighting for cooperative perception in V2V communications,” *IEEE Trans. Intell. Veh.*, vol. 9, no. 2, pp. 3569–3580, 2024, doi: 10.1109/TIV.2023.3345035.
- [3] Y. Yan, Y. Mao, and B. Li, “SECOND: Sparsely embedded convolutional detection,” *Sensors*, vol. 18, no. 10, p. 3337, 2018.
- [4] Y. Hu, S. Fang, Z. Lei, Y. Zhong, and S. Chen, “Where2comm: Communication-efficient collaborative perception via spatial confidence maps,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 4874–4886.
- [5] L. Breiman, “Random forests,” *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

field of view the perfect-channel feature ceiling is below fixed L on test.

Relation to the frozen track. The two tracks share a ground-truth denominator and answer different questions. The deployed track asks what each branch delivers as configured, which is the question a deployment faces and is what every headline number in the main paper reports. The common-volume track asks how much of the L – F gap survives when the field of view is held equal, which is the question a reader asking “is this a granularity effect?” is really asking. Neither supersedes the other, and the diagnostic changes no number in the main paper. It does bound one claim: on test, the headroom a granularity policy could contest is a field-of-view effect.

TABLE XI

Channel-averaged comparison on the frozen OPV2V test split (selector trained on validate, not retrained; $T = 2,170$ frames), read from the frozen replay: F1 and channel use are the means over the same 200 paired CSI samplings (SNR $\sim \mathcal{U}[0, 20]$ dB, 50/50 AWGN/Rayleigh) for every row. The comparison is budget-indexed: each frozen selector is shown against the SNR-threshold rule tuned to the same budget, and it does not run one way. Share is channel use as a fraction of Fixed F (0.990 Msym).

Policy	Mean F1	Channel use (Msym)	Share of Fixed F
Fixed L (object-level)	0.8910	0.024	2.4%
Fixed F (LDPC+16-QAM)	0.8483	0.990	100.0%
Fixed C_{256} (LDPC+256-QAM)	0.8255	0.495	50.0%
Masked oracle (upper bound)	0.9056	0.175	17.7%
$B_{\max} = 0.10$			
SNR-threshold ($\tau^*=18$)	0.89247	0.0724	7.3%
CA-TOSG (RF, frozen)	0.89148	0.0368	3.7%
$B_{\max} = 0.20$			
SNR-threshold ($\tau^*=12$)	0.89701	0.2168	21.9%
CA-TOSG (RF, frozen)	0.89691	0.1414	14.3%
$B_{\max} = 0.30$			
SNR-threshold ($\tau^*=8$)	0.89900	0.3125	31.6%
CA-TOSG (RF, frozen)	0.89783	0.2120	21.4%